Mohamed Khalil **KHLIFI**

Mechatronics Engineer | Embedded Systems & Firmware

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SUMMARY

Mechatronics Engineer specializing in embedded systems, firmware development, and hardware integration. Experienced in microcontroller programming, communication protocols, motor control, and developing complete embedded solutions from electronic design to functional prototypes.

PROFESSIONAL EXPERIENCE

Neuralbey — End of Studies Internship: Smart Pet Feeder — View Source Code | March 2025 – September 2025

- Architected an IoT-based smart pet feeder featuring automated dispensing, remote monitoring and local user interface and safety features.
- Programmed the ESP32 firmware to manage four sensors, an RTC module, stepper motor control and Blynk-based mobile communication.
- Incorporated Wi-Fi connectivity through the ESP32 to enable remote monitoring and scheduled feeding via the Blynk IoT platform.
- Laid out a custom PCB with power management circuitry and hardware interfaces for reliable system integration.

Key Technologies: ESP32 – Embedded C – IoT – Arduino IDE – PCB Design – Altium Designer – Wokwi – SolidWorks – Blynk – Sensors

TAC–TIC — Summer Internship: AI Object Counting System — View Source Code | July 2024 – September 2024

- Created an AI-powered object-counting application capable of processing images, videos, and live webcam streams using YOLOv8.
- Designed a graphical user interface using Tkinter for mode selection and input-source management.

Key Technologies: Python – PyTorch – OpenCV – YOLOv8 – Numpy – PyCharm – Ultralytics – Tkinter – Computer Vision

Relays and Industrial Automation (RAI) — Summer Internship | July 2023 – August 2023

- Analyzed industrial control schematics and supported component selection, electrical assembly, and PCB integration according to system requirements.
- Performed functional testing and troubleshooting to validate hardware operation and ensure system reliability.

Key Technologies: Industrial Automation – Electrical Schematics – Assembly – Functional Testing – PCB Assembly – Troubleshooting

PROJECTS

Stewart Platform (6-DOF Parallel Robot) — View Source Code | March 2026

- Developed ESP32-based control for a 6-DOF Stewart Platform, coordinating six synchronized linear actuators with a 12 V / 10 A power architecture.
- Programmed IMU-based orientation compensation using MPU6050 feedback for real-time platform stabilization.

Key Technologies: ESP32 – MPU6050 – FIT0805 – Robot Kinematics – Control Systems – Parallel Robotics – MATLAB – Electrical Architecture

End of Year Project: Surveillance Robot — View Source Code | February 2024 – May 2024

- Developed an ESP32-CAM-based system for wireless video streaming to a host computer running YOLOv3-based object detection and tracking.
- Enhanced the surveillance system with an existing mobile streaming application support for remote use.

Key Technologies: ESP32-CAM – Arduino IDE – SolidWorks – Fritzing – YOLOv3 – Robotics

Fighter Robot — View Source Code | March 2023

- Built a combat robot controlled by an STM32, integrating a 36 V / 4.5 A drivetrain for high-current BLDC motor operation.
- Flashed a hoverboard motherboard via UART to enable custom motion control, with RC receiver integration for responsive and reliable robot control.

Key Technologies: STM32F103RCT6 – STM32Cube Programmer – VS Code – BLDC Motor Control – RC Communication – Embedded C – Material Integration

All Terrain Robot & Line-Following Robot — View Source Code | February 2022 – May 2022

- Assembled an all-terrain robot using the ESP32 and BTS7960 motor drivers for high-power DC motor control, with PS3 Bluetooth wireless operation.
- Fabricated a line-following robot with five IR sensors and a tuned PD control algorithm for stable, real-time trajectory tracking.

Key Technologies: ESP32 – BTS7960 – Bluetooth – PS3 Controller – PWM – Motor Driver Integration – Arduino – IR Sensors – PD Control – Proteus

EDUCATION

National Engineering School of Carthage (ENICarthage) | September 2021 – October 2025

- National Engineering Degree in Mechatronics Engineering (Master's equivalent)

Preparatory Institute for Engineering Studies of Sfax (IPEIS) | September 2019 – June 2021

- Engineering Preparatory Cycle – Physics and Technology

TECHNICAL SKILLS

Embedded Systems: Firmware Development – STM32 Programming – ESP32 / Arduino Development – Interrupts – Motor Control – ADC – Timers – PWM
Communication Interfaces: UART – I^2C – SPI – CAN – Bluetooth – Wi-Fi
Electronics & PCB Design: PCB Design – Electrical Schematics – Power Electronics – Sensor Integration – Motor Driver Integration – Hardware Validation
Robotics & Control Systems: Robot Design – Control Systems – Motion Control – PID/PD Control – Kinematics – System Integration
Debugging & Validation: Oscilloscope – Digital Multimeter – Circuit Debugging – Hardware Troubleshooting
Design Tools: Altium Designer – Proteus – SolidWorks
Programming Languages: Embedded C – C – Python
Development Tools: STM32CubeIDE – STM32CubeMX – Arduino IDE – Git – GitHub – VS Code

CERTIFICATIONS & TRAINING

Microcontroller Embedded C Programming: Foundations for Firmware Engineers – FastBit Embedded Brain Academy **| June 2026**
Key Topics: Cortex-M – Embedded C – STM32 – Memory-Mapped Registers – Interrupts – Bit Manipulation – STM32CubeIDE

LANGUAGES

Arabic: Native proficiency **| English:** Professional Working Proficiency **| French:** Intermediate proficiency

ASSOCIATIVE ACTIVITIES

IEEE ENICarthage Student Branch / Mechatronics ENICarthage Club – Member **| October 2021 – December 2024**
ENICarthage Robots – Head of Line Following Committee **| February 2022**
ENICarthage Internal Contest, Line-Following Competition – 1st Place Winner **| January 2022**